

HEALTHCARE ANALYTICS DASHBOARD

Week 6 Task Report

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BSC DATA SCIENCE & ARTIFICIAL INTELLIGENCE

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1. Introduction & Objective

This report documents the Week 6 task: building an end-to-end Healthcare Analytics dashboard in Power BI on top of a hospital admissions and billing dataset. The work covers data modelling, DAX measure development, visual layout planning, and final dashboard assembly.

The goal of the dashboard is to give hospital administrators and finance teams a single view of patient volumes, billing performance, average length of stay, and readmission trends, filterable by insurance provider, admission type, and date range.

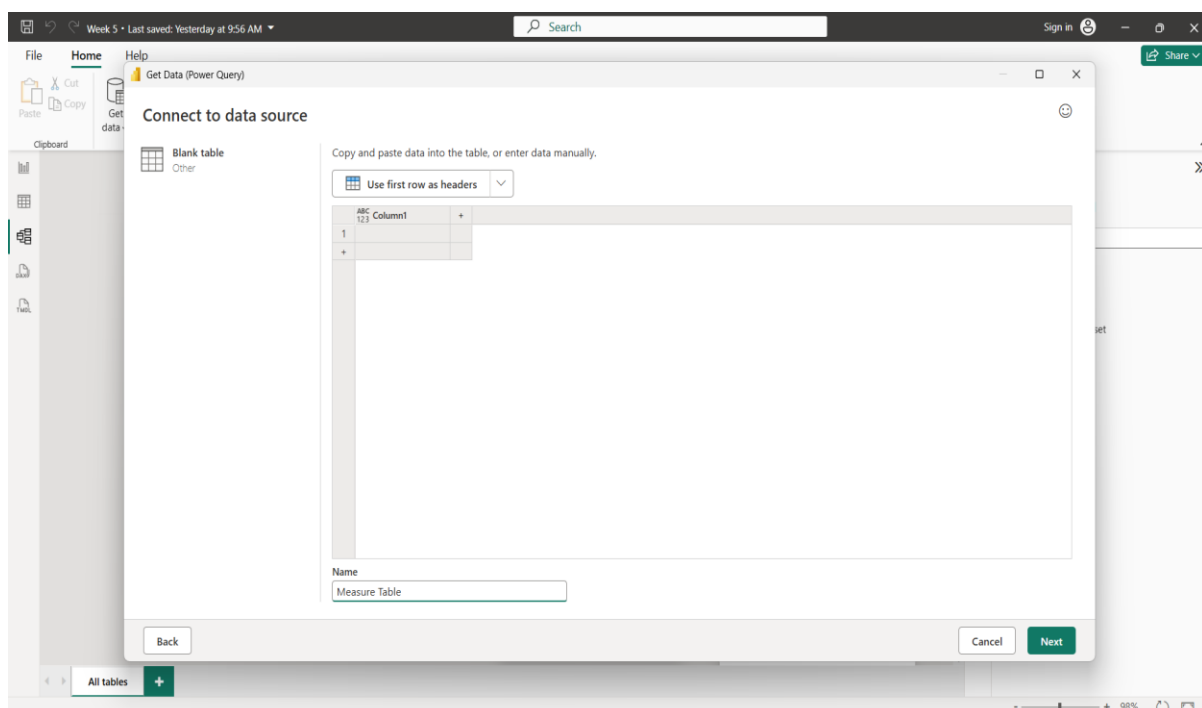
1.1 Dataset Overview

The source data (healthcare_dataset) contains one row per billing record, with the following key attributes:

- Patient details: Name, Age, Age Group, Gender, Blood Type
- Admission details: Admission ID, Admission Type, Date of Admission, Discharge Date, Doctor, Hospital, Room Number
- Clinical details: Medical Condition, Medication, Test Results
- Financial details: Billing Amount, Insurance Provider

2. Data Model Setup

Before writing any DAX, a dedicated Measure Table was created to keep all calculated measures separate from the raw data tables. This was done using Power BI's "Enter Data" (Get Data → Blank table) option, giving a clean home for every measure created in Section 4.



The final model is a star schema with a central Billing fact table connected to Dim_patient and Dim_Admission dimension tables, plus the original flat healthcare_dataset kept for reference and the standalone Measure Table:

Table	Role
Billing	Fact table — Admission ID, Billing Amount, Date of Admission, Discharge Date, Insurance Provider, Patient ID
Dim_patient	Dimension — Age, Age Group, Blood Type, Gender, Name, Patient ID
Dim_Admission	Dimension — Admission ID, Admission Type, Doctor, Hospital, Medical Condition, Medication, Room Number, Test Results
Measure Table	Houses all calculated DAX measures (no data columns)
healthcare_dataset	Original flat source table, retained for validation

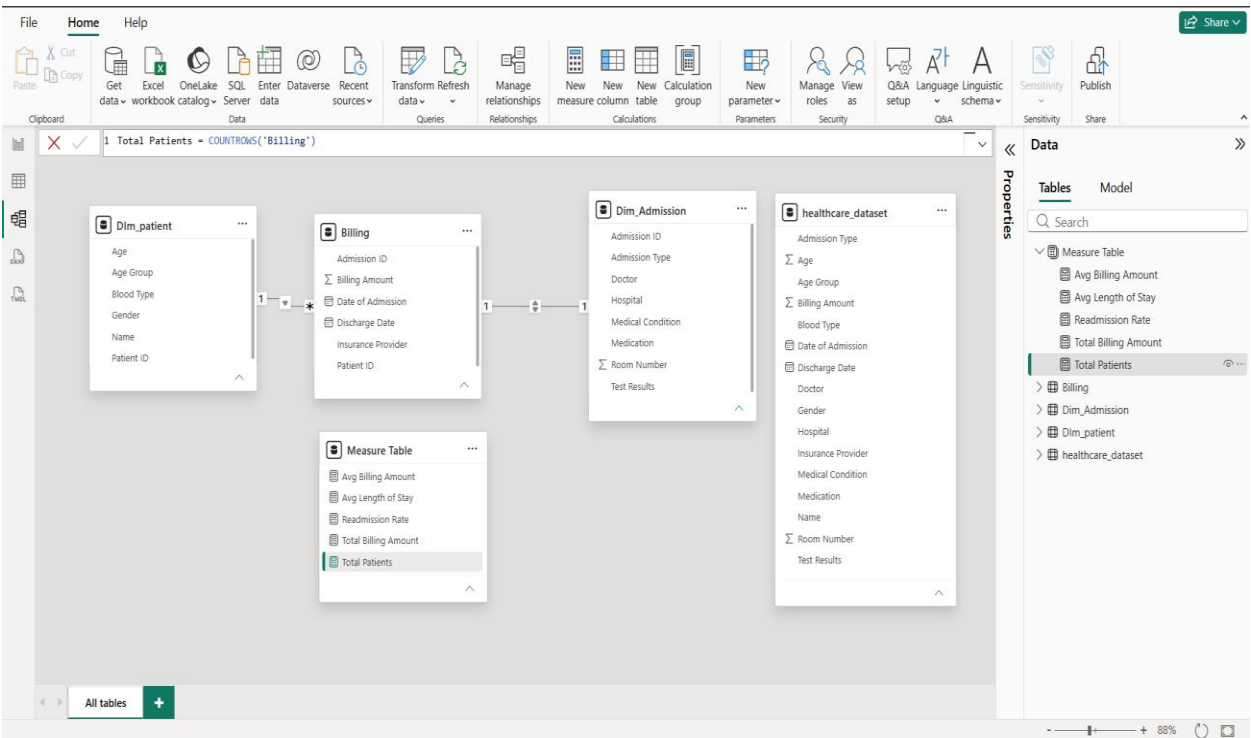
3. DAX Measures Development

Five core measures were planned and then implemented against the model. Each measure was added to the Measure Table one at a time and verified in the Model view before moving to the next.

3.1 Total Patients

Counts the number of billing/admission records, used as the base patient-volume KPI.

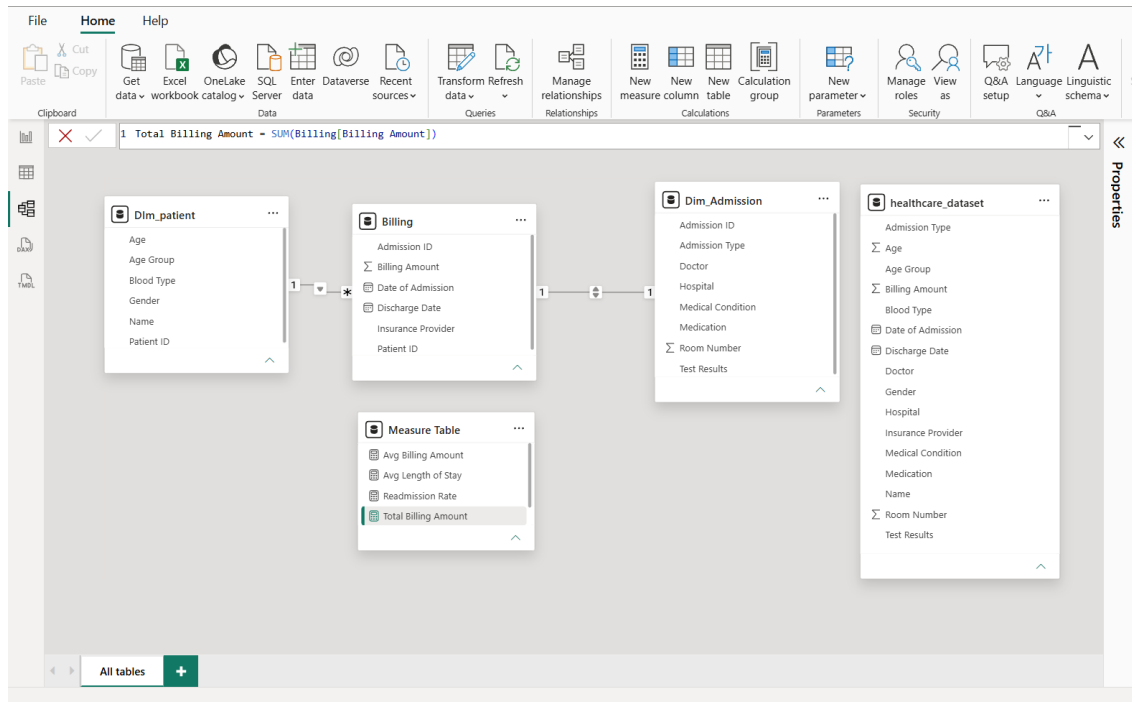
Total Patients = COUNTROWS(Billing)



3.2 Total Billing Amount

Sums the billing amount across all records to give total healthcare revenue.

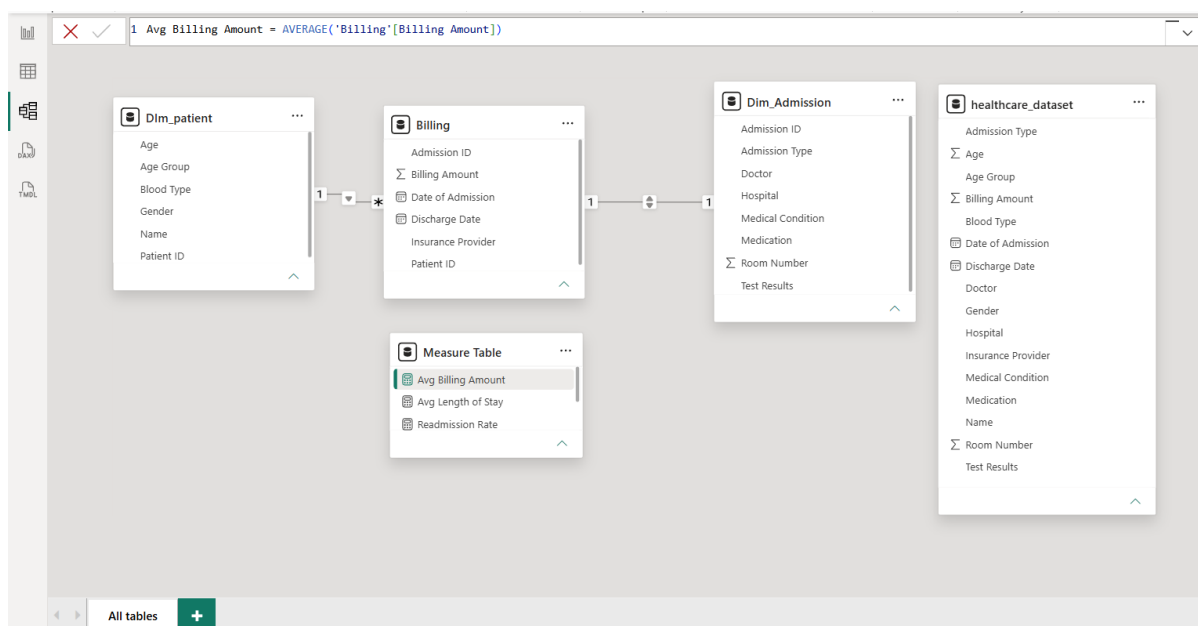
```
Total Billing Amount = SUM(Billing[Billing Amount])
```



3.3 Average Billing Amount per Patient

Gives the average cost per billing record, useful for cost-per-patient benchmarking.

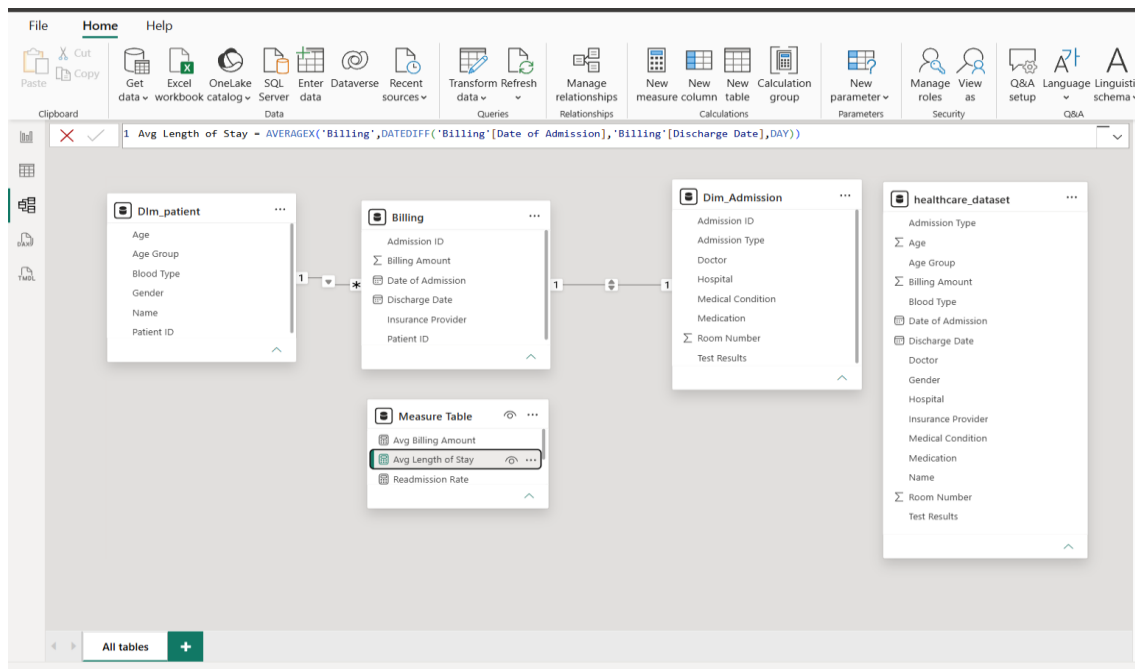
```
Avg Billing Amount = AVERAGE('Billing'[Billing Amount])
```



3.4 Average Length of Stay (Days)

Calculates the average number of days between admission and discharge for each record using DATEDIFF inside AVERAGEX.

```
Avg Length of Stay =  
AVERAGEX(  
    Billing,  
    DATEDIFF(Billing[Date of Admission], Billing[Discharge Date], DAY)  
)
```



3.5 Readmission Rate

Identifies patients who appear more than once in the Billing table (repeat admissions) and expresses them as a share of total patients, using SUMMARIZE + FILTER to build a per-patient admission count, then DIVIDE for a safe ratio.

```
Readmission Rate =  
VAR TotalCount = [Total Patients]  
VAR RepeatPatients =  
    COUNTROWS(  
        FILTER(  
            SUMMARIZE(  
                Billing,  
                Billing[Patient ID],  
                "Admission Count", COUNTROWS(Billing)  
            ),  
            [Admission Count] > 1  
        )  
    )  
RETURN  
    DIVIDE(RepeatPatients, TotalCount, 0)
```

File Home Help

Clipboard: Paste, Cut, Copy

Data: Get data, Excel workbook catalog, OneLake Server, SQL Enter data, Dataserve Recent sources, Transform Refresh data, Queries, Manage relationships, Relationships, New measure column, Calculations, New table, Calculation group, New parameter, Parameters, Manage roles, View as, Security, Q&A Language setup, Linguistic schema, Q&A, Sensitivity, Publish, Share

1 Readmission Rate = VAR TotalCount = [Total Patients]
 2 VAR RepeatPatients = COUNTROWS(FILTER(SUMMARIZE('Billing','Billing'[Patient ID],'Admission Count',COUNTROWS('Billing')),[Admission Count] > 1))
 3 RETURN DIVIDE(RepeatPatients,TotalCount,0)

Dim_patient: Age, Age Group, Blood Type, Gender, Name, Patient ID

Billing: Admission ID, Billing Amount, Date of Admission, Discharge Date, Insurance Provider, Patient ID

Dim_Admission: Admission ID, Admission Type, Doctor, Hospital, Medical Condition, Medication, Room Number, Test Results

healthcare_dataset: Admission Type, Age, Age Group, Billing Amount, Blood Type, Date of Admission, Discharge Date, Doctor, Gender, Hospital, Insurance Provider, Medical Condition, Medication, Name, Room Number, Test Results

Measure Table: Avg Billing Amount, Avg Length of Stay, Readmission Rate, Total Billing Amount, Total Patients

Properties: Tables, Model, Search, Measure Table, Avg Billing Amount, Avg Length of Stay, Readmission Rate, Total Billing Amount, Total Patients, Billing, Dim_Admission, Dim_patient, healthcare_dataset

All tables +

88%

3.6 All Measures

File Home Help

Clipboard: Paste, Cut, Copy

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Dim_patient: Age, Age Group, Blood Type, Gender, Name, Patient ID

Billing: Admission ID, Billing Amount, Date of Admission, Discharge Date, Insurance Provider, Patient ID

Dim_Admission: Admission ID, Admission Type, Doctor, Hospital, Medical Condition, Medication, Room Number, Test Results

healthcare_dataset: Admission Type, Age, Age Group, Billing Amount, Blood Type, Date of Admission, Discharge Date, Doctor, Gender, Hospital, Insurance Provider, Medical Condition, Medication, Name, Room Number, Test Results

Measure Table: Avg Billing Amount, Avg Length of Stay, Readmission Rate, Total Billing Amount, Total Patients

Properties: Tables, Model, Search, Measure Table, Avg Billing Amount, Avg Length of Stay, Readmission Rate, Total Billing Amount, Total Patients, Billing, Dim_Admission, Dim_patient, healthcare_dataset

All tables +

88%

4. Visual Layout & Dashboard Design

With the measures in place, the dashboard layout was planned before building, so every visual has a clear purpose and data-field mapping. The plan below was used as the build checklist for Section 5.

4.1 Top KPI Summary Cards

- Total Patients
- Total Billing Amount
- Average Billing Amount
- Average Length of Stay

4.2 Demographics & Patient Distribution

Matrix visual — Rows: Age Group, Gender | Columns: Medical Condition | Values: Total Patients, Average Billing Amount.

Donut chart — Legend: Age Group | Values: Total Patients.

4.3 Medical Conditions & Admission Volumes

Stacked bar chart / matrix — Rows: Medical Condition | Values: Total Patients, split by Admission Type.

4.4 Admission Volume Trends & Hospital Analytics

Line chart — X-axis: Date of Admission (Year, Month) | Y-axis: Total Patients, Total Billing Amount.

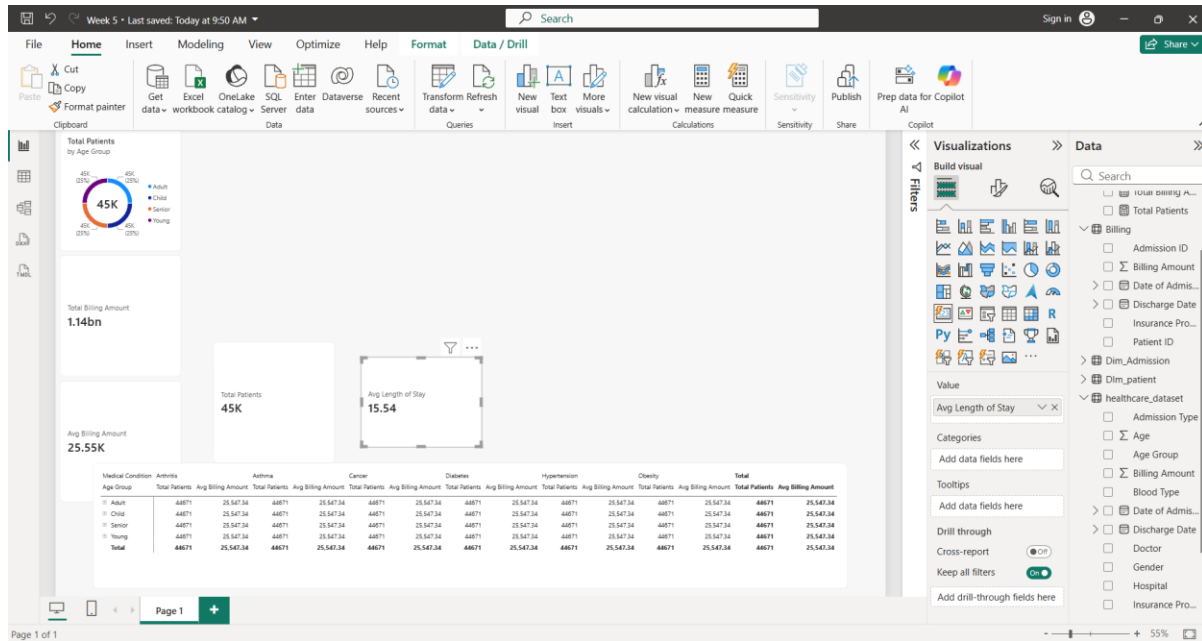
Clustered column chart — X-axis: Insurance Provider | Y-axis: Total Billing Amount.

4.5 Slicers & Dynamic Cross-Filtering

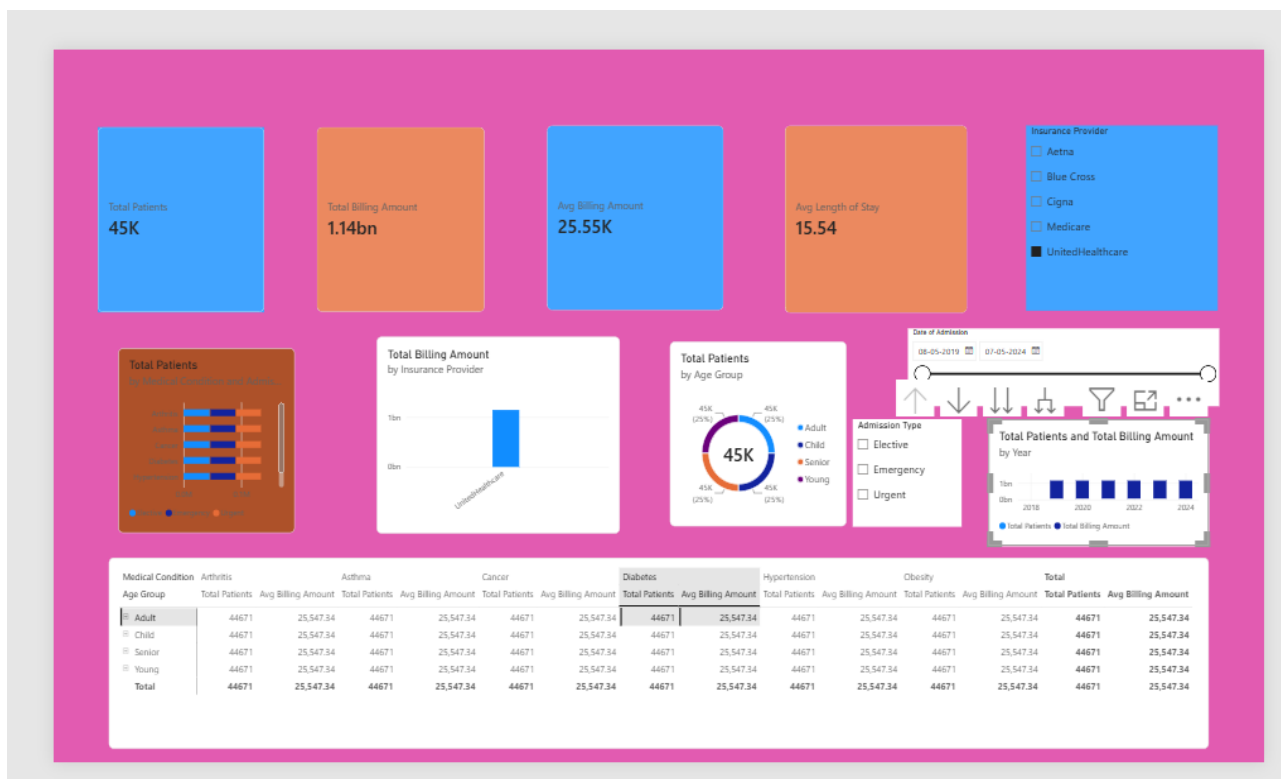
- Slicer 1 (Tile/Dropdown): Insurance Provider
- Slicer 2 (Tile/Dropdown): Admission Type
- Slicer 3 (Date Range): Date of Admission

5. Report Building Process

The visuals from the layout plan were added to the report canvas one at a time, using the Fields pane to bind Total Patients, Avg Billing Amount, and Avg Length of Stay to cards, a matrix, and a donut chart.

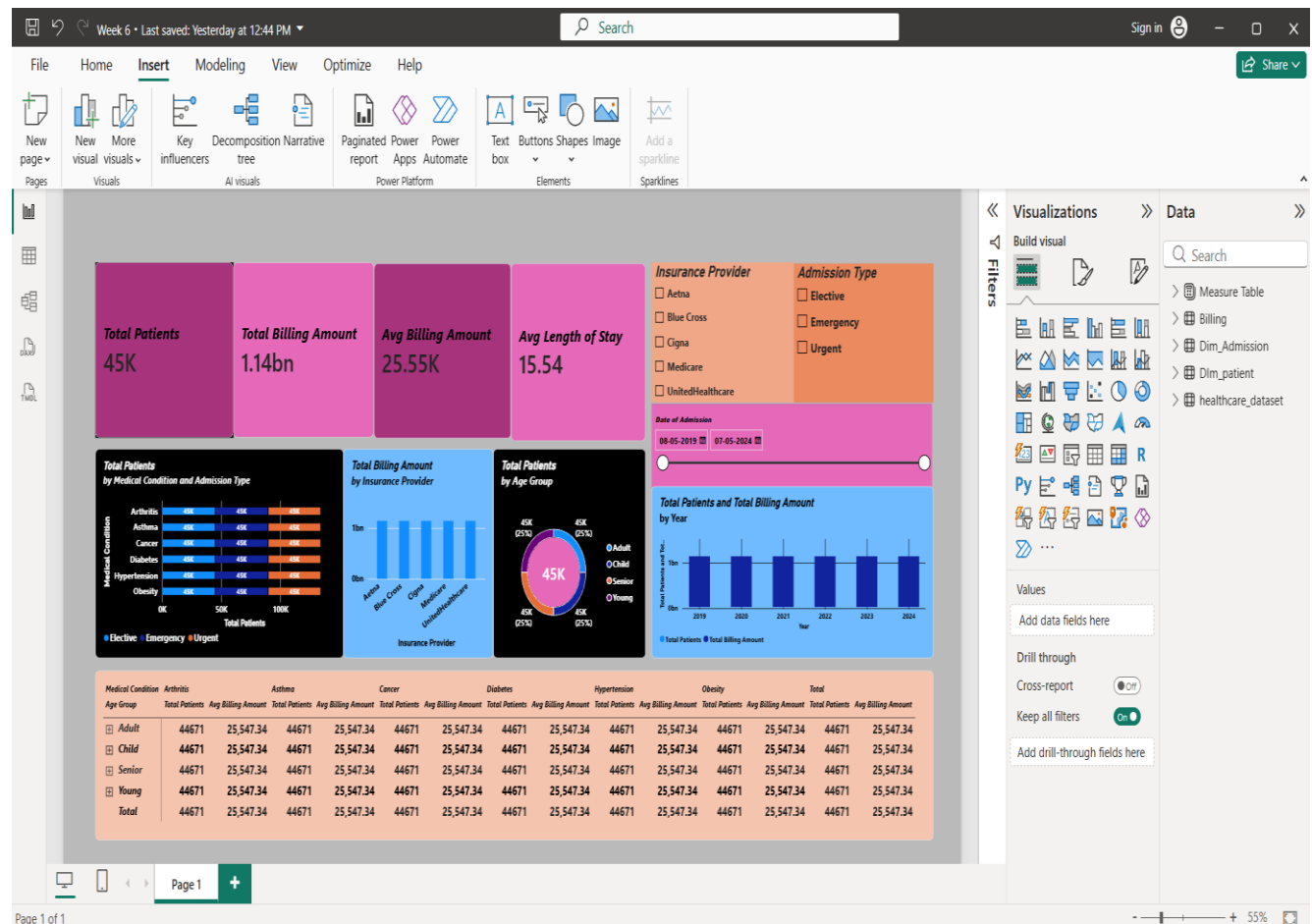


A wireframe pass was then used to lay out every planned visual — KPI cards, the Total Patients donut, the Total Billing Amount column chart, Insurance Provider and Admission Type slicers, the Date of Admission range slicer, the Total Patients/Total Billing Amount trend chart, and the full demographics matrix — before final formatting.



6. Final Dashboard

The completed dashboard brings together every planned element: four KPI cards at the top (Total Patients, Total Billing Amount, Avg Billing Amount, Avg Length of Stay), medical-condition and insurance-provider breakdowns, an admissions trend chart, cross-filtering slicers, and the full age-group / medical-condition matrix at the bottom.



All visuals respond to the Insurance Provider, Admission Type, and Date of Admission slicers, giving a single interactive view for exploring patient volume, cost, and readmission patterns across the hospital network.

7. Conclusion

This task walked through the full Power BI workflow: designing a star-schema data model, isolating calculations in a dedicated Measure Table, writing and validating five DAX measures (including a readmission-rate calculation using SUMMARIZE and FILTER), planning a visual layout against business questions, and assembling those visuals into a single interactive dashboard.